ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Low oxygen requires fast breathing.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Spinal cord reflex centre.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Copying errors produce small variations, which explains the statement.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Fusion with X or Y sperm decides the sex.
5	**(c)** 6	$C_6H_{12}O_6 + 6\,O_2 \longrightarrow 6\,CO_2 + 6\,H_2O$.
6	**(c)** diabetes	Insulin.
7	**(d)** variation, and those from the cutting are identical to the parent	Sexual gives variation; vegetative does not.
8	**(b)** 50	1/4 of 200 = 50 (Rryy).
9	**(a)** uses carbon dioxide and releases oxygen	Photosynthesis replenishes oxygen.
10	**(a)** adrenaline	Fight or flight.
11	**(b)** undifferentiated cells to produce missing parts	Specialised dividing cells are essential.
12	**(b)** 1 : 1 : 1 : 1	Four gamete types x one.
13	**(a)** Both A and R are true and R is the correct explanation of A.	Valves prevent backflow.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true but not an explanation of neuron direction itself.
15	**(d)** A is false but R is true.	A is false: most organisms cannot reproduce by regeneration; R is true.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Volume per minute = 72 x 70 (1); = 5040 mL (5.04 L) (1).	1+1
17	Motor neuron or effector (1); Sensory path is intact (1).	1+1
18	Regeneration (1); It involves growth of a whole body from a body part, not an outgrowth (1).	1+1
19	TT fraction = 1/4 (1); 60 plants (1).	1+1
20	Bile neutralises acid from the stomach (1); Pancreatic enzymes work in alkaline medium (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Light is needed for photosynthesis (1); Starch is formed in light only (1); Covered region shows no starch (1).	1+1+1
22	Tip makes auxin (1); Auxin goes to the shaded side (1); No tip: no auxin, no bending (1).	1+1+1
23	Divisions in 2 hours = 8 (1); Number = 2^8 = 256 (1); Binary fission (asexual) (1).	1+1+1
24	Cross with green wrinkled (yyrr) (1); All yellow round offspring means YYRR (1); Other phenotypes in offspring mean the plant is heterozygous (1).	1+1+1
	OR: Heart works continuously (1); Needs oxygen and glucose (1); Blockage leads to heart attack (1).	1+1+1
25	Motor nerve carries signal to muscle (1); Effector is the muscle (1); Sensory part not needed (1).	1+1+1
	OR: Plants have meristematic cells for new growth (1); Humans have complex organ systems (1); Human body parts cannot make a whole body (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Aim: to show CO_2 is needed for photosynthesis (1); KOH absorbs CO_2 (1); Leaf with KOH: no starch (1); Leaf with water: starch present (1); Control: leaf in water flask (1).	1+1+1+1+1
	OR: Aim: role of tip in phototropism (1); Take three seedlings: normal, tip cut, tip covered with foil (1); Expose to light from one side (1); Observation: only normal seedling bends (1); Conclusion: tip senses light and makes auxin (1).	1+1+1+1+1
27	Asexual offspring are nearly identical clones (1); DNA copying errors give small variations only (1); Sexual reproduction mixes DNA of two parents (1); Variations help some survive changes in the environment (1); Example: cross-pollination in plants gives varied seeds (1).	1+1+1+1+1
	OR: F1: TtRr, tall round (1); F2 ratio 9 : 3 : 3 : 1 (1); Tall round 900, tall wrinkled 300 (1); Dwarf round 300, dwarf wrinkled 100 (1); Dwarf round genotypes ttRR and ttRr (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Germ-cell DNA is unchanged, so acquired traits are not passed on.
2	**(c)** 1	$6\,CO_2 + 6\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2$.
3	**(c)** reflex action is faster than brain sensation	Information reaches the brain later.
4	**(d)** spores need moisture to germinate	Water is needed to activate spores.
5	**(c)** 1/4	Each birth is independent.
6	**(b)** chlorophyll is needed for photosynthesis	Chlorophyll.
7	**(b)** iodine	Goitre.
8	**(a)** shortage of nutrients and oxygen	The placenta is the exchange site.
9	**(b)** new combinations of independently inherited genes	Independent assortment gives new combinations.
10	**(c)** A is true but R is false.	Bile has no enzymes; it emulsifies fats.
11	**(c)** A is true but R is false.	A true; R false, roots grow downward.
12	**(a)** Both A and R are true and R is the correct explanation of A.	The scrotum keeps the testes cooler.
13	**(a)** Both A and R are true and R is the correct explanation of A.	Masking explains absence of the recessive trait.
14	**(b)** 4320	72 x 60.
15	**(b)** feel the touch but not move the muscle	Sensory pathway is intact.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Feedback mechanism (1); Level depends on need (1).	1+1
17	Unisexual flower (1); Pollen must be moved from the male flower to the female flower (1).	1+1
18	T is dominant over t (1); The tall allele masks the dwarf allele (1).	1+1
19	Low haemoglobin (1); Less oxygen to cells, less energy (1).	1+1
20	Adrenaline released (1); Prepares the body for action (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Sperm 23 (22 autosomes) (1); Egg 23 (22 autosomes); zygote 46 (44 autosomes) (1); Liver cell 46 (44 autosomes) (1).	1+1+1
22	Salivary amylase breaks starch to maltose (1); Stomach acid stops amylase (1); Pancreatic amylase in small intestine gives glucose (1).	1+1+1
23	Adrenaline raises sugar for energy (1); Insulin lowers sugar (1); Opposing action (1).	1+1+1
24	Embryo would not implant (1); Pregnancy would not be established (1); Embryo could not be nourished (1).	1+1+1
	OR: A woman has two X chromosomes (XX) (1); Each egg receives only one of them (1); So each egg has one X and 22 autosomes (1).	1+1+1
25	Atmospheric oxygen is lower (1); Faster breathing to get more oxygen (1); More RBC for transport (1).	1+1+1
	OR: Eyes send signals to brain (1); Brain stimulates adrenal glands (1); Adrenaline prepares for flight (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	Aerobic: mitochondria, CO_2 + H_2O (1); Yeast: ethanol + CO_2 (1); Muscle: lactic acid (1); Only 2 ATP in anaerobic (1); Complete oxidation yields about 30 ATP (1).	1+1+1+1+1
	OR: Root bends down (1); Shoot bends up (1); Auxin collects on lower side (1); High auxin promotes shoot growth, inhibits root (1); Gravity-induced response (1).	1+1+1+1+1
27	Egg is released from the ovary into the oviduct (1); Sperm fuses with the egg in the oviduct (1); Zygote divides and moves towards the uterus (1); Embryo is implanted in the thick lining (1); Placenta forms from embryonic and uterine tissue (1).	1+1+1+1+1
	OR: First two plants: dwarf offspring means each has t (1); 1/4 dwarf shows both are Tt (1); Third plant crossed with tt gives all tall (1); So it must be TT (1); If it were Tt, half would be dwarf (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 8	Three divisions give $2^3 = 8$.
2	**(a)** Tt	1 : 1 ratio.
3	**(a)** both plants	No light or no CO2 means no photosynthesis.
4	**(a)** gigantism	Excess growth hormone gives excessive growth.
5	**(a)** The egg cannot meet the sperm	Fertilisation occurs in the oviduct.
6	**(b)** environment affects expression of the trait	Phenotype depends on environment.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Transpiration pull.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Prevents goitre.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Equal gametes T and t give TT : Tt : tt = 1 : 2 : 1.
11	**(c)** 5040	70 x 72.
12	**(a)** not feel a stimulus	Information does not reach CNS.
13	**(b)** 16	$2^4 = 16$.
14	**(a)** 1/4	Four equally likely types.
15	**(d)** more carbon dioxide	Exhaled air has about 4% CO_2.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Pollen grain (1); Formed in the anther (1).	1+1
17	Ff x ff gives 1 : 1 (1); 2 children (1).	1+1
18	Reabsorbed 178.5 L (1); Percentage = 99.2% (1).	1+1
19	Roots grow into the soil (1); Absorb water and minerals (1).	1+1
20	Number of divisions = 6 (1); $2^6 = 64$ Amoebae (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Ovaries release the egg and secrete oestrogen (1); Oestrogen controls thickening of the uterine lining (1); Without it the lining is not formed or shed (1).	1+1+1
22	Each pregnancy is independent (1); X and Y sperm are equally likely (1); Probability = 1/2 (1).	1+1+1
23	Tubule reabsorption is faulty (1); Normally all glucose is reabsorbed (1); Diabetes may be a cause (1).	1+1+1
24	Chemicals from ovule attract (1); Chemotropism (1); Growth of pollen tube (1).	1+1+1
	OR: Pollen tube must grow a long distance (1); It grows through the style tissue (1); It delivers male gametes to the ovule (1).	1+1+1
25	Green beetles are less easily seen (1); They survive and reproduce more (1); The proportion of green beetles increases (1).	1+1+1
	OR: Soil solution is more dilute (1); Water moves into root hair cell (1); Moves into xylem (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	Per minute: 7500 mL (1); Per hour: 450000 mL = 450 L (1); Oxygen = 21% of 450 = 94.5 L (1); Not all oxygen is absorbed (1); Exhaled air has about 16% oxygen (1).	1+1+1+1+1
	OR: Receptor senses heat (1); Sensory neuron to spinal cord (1); Motor neuron to muscle (1); Hand withdrawn (1); Brain feels pain later and the person screams (1).	1+1+1+1+1
27	Moist and warm: rapid growth of mould (1); Dry: little or no growth (1); Moist and cold: slow growth (1); Spores germinate only with moisture and warmth (1); Thick-walled spores wait for suitable conditions (1).	1+1+1+1+1
	OR: Tt x tt gives 1 tall : 1 dwarf (1); Rr x Rr gives 3 round : 1 wrinkled (1); Combine the independent ratios (1); Tall round 3, tall wrinkled 1, dwarf round 3, dwarf wrinkled 1 (1); Ratio 3 : 1 : 3 : 1 (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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